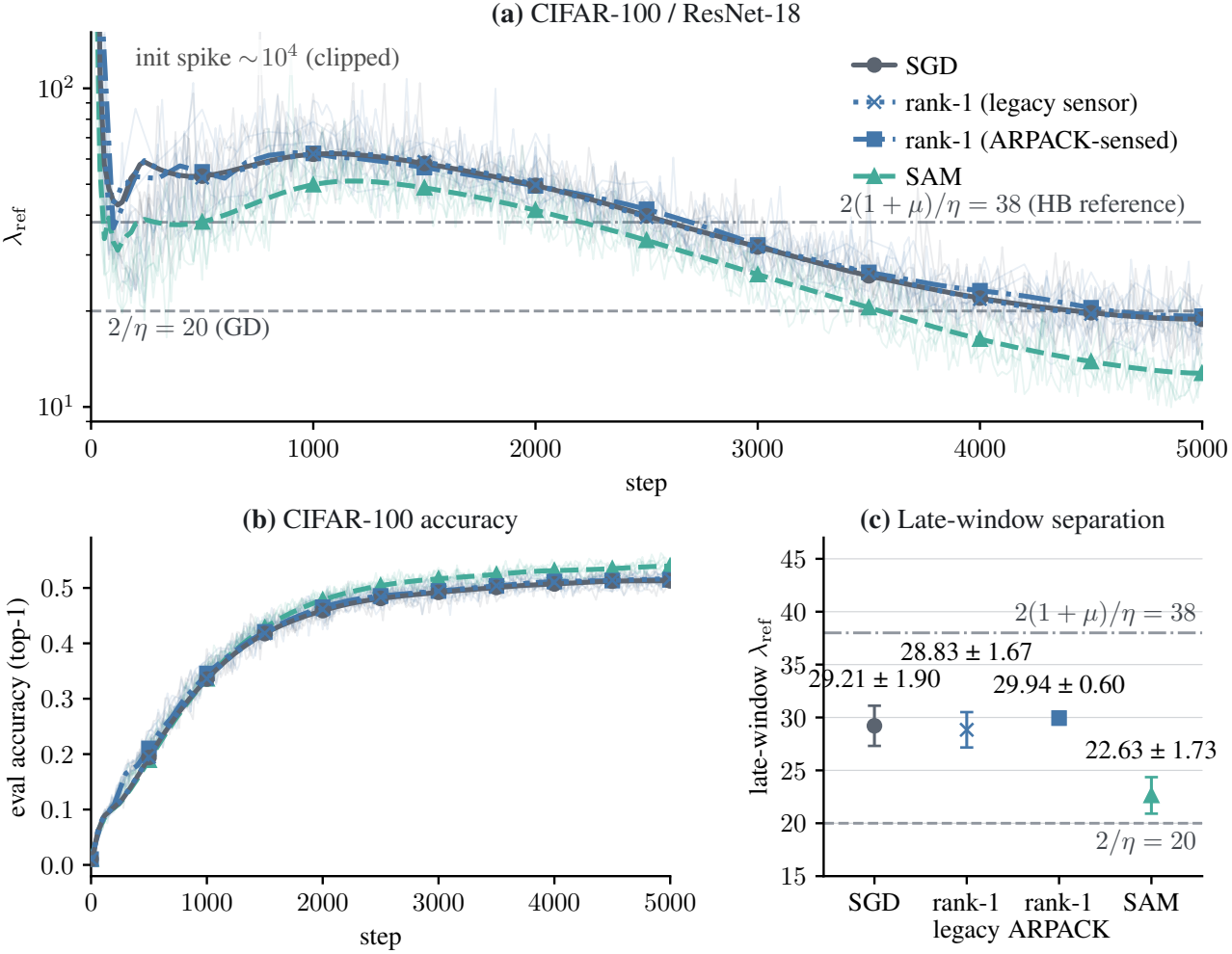

(a) CIFAR-100 / ResNet-18
init spike ∼10^4 (clipped)
SGD
rank-1 (legacy sensor)
rank-1 (ARPACK-sensed)
SAM
2(1 + μ)/η = 38 (HB reference)
2/η = 20 (GD)
λref
step
(b) CIFAR-100 accuracy
eval accuracy (top-1)
step
(c) Late-window separation
late-window λref
2(1 + μ)/η = 38
28.83 ± 1.67
29.21 ± 1.90
29.94 ± 0.60
22.63 ± 1.73
2/η = 20
SGD
rank-1 legacy
rank-1 ARPACK
SAM